SIMATS ENGINEERING

ASSESSMENT TOOL 2 - CONCEPT MAPPING

Machine Learning - ITA06

Course Outcome: CO1 AT2

Topic: Concept Mapping, Find-S and Candidate Elimination

Complete Answer with Sketches and Step-by-Step Hypothesis Updates

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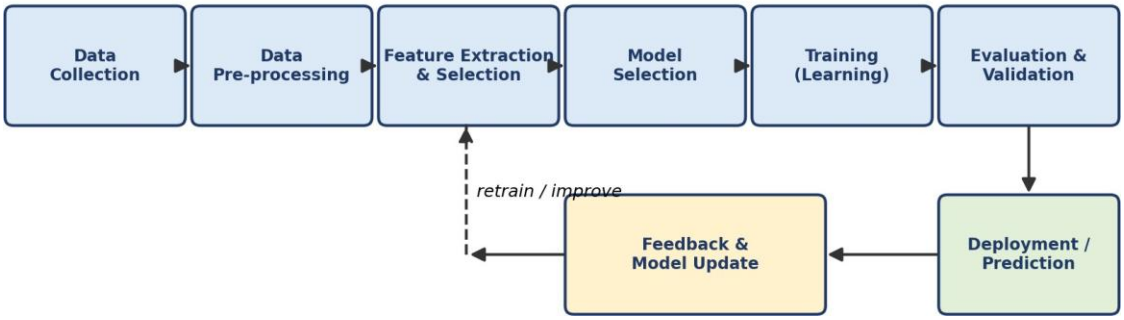
Machine Learning - ITA0602

Slot: C

Question 1: Blocks of a Machine-Learning System

A machine-learning (ML) system converts raw data into a model that can make decisions on unseen data. It consists of the following blocks, shown in Fig. 1.

Fig. 1 Blocks of a Machine-Learning System



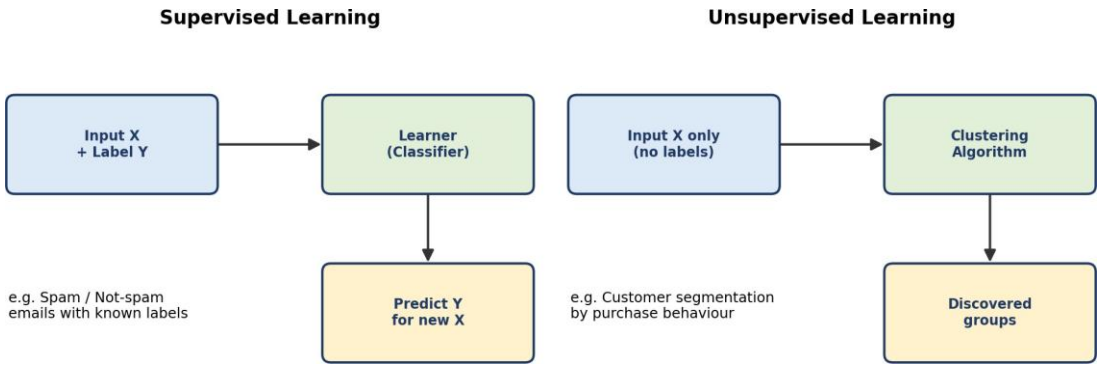
Explanation of blocks

- **1. Data Collection:** Gathering raw data from sensors, databases, logs, surveys or the web. Quality and quantity of data decide the ceiling of model performance.
- **2. Data Pre-processing:** Cleaning noise, handling missing values, removing duplicates, normalising/scaling, encoding categorical values.
- **3. Feature Extraction & Selection:** Extracting useful attributes from raw data and selecting the most relevant ones (e.g. PCA, correlation, information gain) to reduce dimensionality and overfitting.
- **4. Model Selection:** Choosing a suitable hypothesis space/algorithm (decision tree, SVM, neural network, k-means, etc.) according to the problem and the inductive bias.
- **5. Training (Learning):** The learning algorithm adjusts model parameters using the training set to minimise an error/loss function.

- **6. Evaluation & Validation:** Performance is checked on validation/test data using metrics such as accuracy, precision, recall, F1-score and RMSE; hyper-parameters are tuned.

- **7. Deployment / Prediction:** The trained model predicts outputs for new, unseen inputs in a real application.
- **8. Feedback & Model Update:** Prediction errors and new data are fed back so the model can be retrained and improved.

Supervised vs. Unsupervised Learning



Aspect	Supervised Learning	Unsupervised Learning
Training data	Labelled (input X with known output Y)	Unlabelled (only input X)
Goal	Learn mapping X → Y to predict outputs	Discover hidden structure, groups or patterns
Typical tasks	Classification, regression	Clustering, association, dimensionality reduction
Example	Spam filter: emails labelled "spam / not spam" are used to train a classifier that labels new emails.	Customer segmentation: an online store groups customers by purchase behaviour using k-means without any

	House-price prediction from area, location (regression).	predefined labels.
Algorithms	Decision tree, Naive Bayes, SVM, linear regression	k-means, hierarchical clustering, PCA, Apriori

Question 2: ML System for Disaster Management

Objective: predict floods and earthquakes using historical records and real-time sensor feeds, and trigger early warnings.

System design

- **Data sources:** Historical: rainfall, river discharge, soil moisture, past flood maps, seismic catalogues (magnitude, depth, location), fault-line data. Real-time: rain gauges, river-level and water-flow sensors, satellite/radar imagery, seismometers, weather forecasts, IoT streams.
- **Data pre-processing:** Handle missing sensor readings (interpolation/imputation), remove outliers and sensor faults, normalise units and scales, align time stamps, resample to uniform intervals, convert imagery to grids, and balance the rare-event class using SMOTE or class weights (disasters are rare).
- **Feature selection:** Rolling statistics (24-hr/72-hr cumulative rainfall), rate of change of river level, soil saturation, elevation/slope, distance to river, seismic energy release and foreshock counts. Selection by correlation analysis, mutual information/information gain, Random-Forest feature importance, and PCA for dimensionality reduction.
- **Choice of model:**
- Floods – Random Forest / Gradient Boosting for tabular data, and LSTM (time-series) for river-level forecasting; CNN on satellite images for flood-extent mapping. Earthquakes – LSTM/CNN-LSTM or Random Forest on seismic sequences to estimate probability/magnitude of events. Ensembles are preferred because they are robust to noise and give feature importance. A Decision Tree is kept as an interpretable baseline.

(a) Training, Validation and Evaluation

- **Split:** Split data in time order (e.g. 70% train, 15% validation, 15% test) – never shuffle time-series data, to avoid leakage.

- **Training:** Fit the model on the training set by minimising a loss (cross-entropy for event/no-event, MSE for water level); use class weights for rare events.
- **Validation:** Tune hyper-parameters (tree depth, learning rate, LSTM units) with time-series cross-validation; use early stopping to prevent overfitting.

- **Evaluation:** On the unseen test set use Precision, Recall (most important – a missed disaster is costly), F1-score, ROC-AUC, confusion matrix, false-alarm rate and lead time for prediction; use RMSE/MAE for water-level forecasts.
- **Testing & Maintenance:** Back-test on past disasters, then retrain periodically as new data arrive.

(b) Early Warning and Emergency Response

- **Real-time monitoring:** The streaming data are pre-processed continuously; the model outputs a risk probability and risk level (Low / Moderate / High / Severe) for each region.
- **Early warning:** When the probability crosses a threshold, alerts are automatically sent by SMS, mobile apps, sirens, radio/TV and social media to authorities and residents, giving precious lead time.
- **Emergency response planning:** Risk maps (GIS) identify vulnerable zones, safe evacuation routes, shelters and hospitals; resources (rescue teams, boats, food, medicines) are pre-positioned.
- **Feedback loop:** Post-event data (actual outcomes, damage reports) are fed back to improve the model and reduce false alarms.

Question 3: Find-S Algorithm – Play Tennis

Hypothesis form: ⟨Outlook, Temperature, Humidity, Wind⟩; "∅ " = no value accepted, "?" = any value accepted. Find-S starts with the most specific hypothesis and generalises it only as far as needed to cover each positive example.

Step 1 – Identify examples: Positive (Yes): Ex 3, 4, 5, 7. Negative (No): Ex 1, 2, 6, 8 → ignored.

Initial hypothesis: h0 = ⟨∅, ∅ , ∅ , ∅ ⟩

Step	Positive example	Hypothesis after step	Reasoning
Ex 1, 2	Negative – ignored	h0 = ⟨∅,∅,∅,∅⟩	No change
Ex 3	⟨Overcast, Hot, High, Weak⟩	h1 = ⟨Overcast, Hot, High, Weak⟩	First positive: every ∅ replaced by the example's value
Ex 4	⟨Rainy, Mild, High, Weak⟩	h2 = ⟨?, ?, High, Weak⟩	Overcast≠Rainy → ?; Hot≠Mild → ?; High, Weak match
Ex 5	⟨Rainy, Cool, Normal, Weak⟩	h3 = ⟨?, ?, ?, Weak⟩	High≠Normal → ?; Weak matches
Ex 6	Negative – ignored	h3 = ⟨?,?,?,Weak⟩	No change
Ex 7	⟨Overcast, Mild, Normal, Strong⟩	h4 = ⟨?, ?, ?, ?⟩	Weak≠Strong → ?
Ex 8	Negative – ignored	h4 = ⟨?,?,?,?⟩	No change

Final Answer: The maximally specific hypothesis is h = ⟨?, ?, ?, ?⟩ (every attribute value is accepted).

Remark: The final hypothesis classifies every day as "Yes", so it wrongly covers the negative examples 1, 2, 6 and 8. This shows that the positive examples cannot be described by a single

conjunctive hypothesis and that Find-S, which ignores negative examples, cannot detect such inconsistency or noise in the data.

Question 4: Candidate Elimination Algorithm – Enjoy Sport

Attributes: ⟨Weather, Temperature, Humidity, Wind⟩. **Initial boundaries:** S0 = {⟨∅,∅,∅,∅⟩}, G0 = {⟨?,?,?,?⟩}

Rules: Positive example → remove from G any hypothesis that does not cover it, and minimally generalise S. Negative example → remove from S any hypothesis that covers it, and minimally specialise G (keeping only those more general than some member of S).

Ex	Training example	S boundary	G boundary
0	Initial	S0 = {⟨∅,∅,∅,∅⟩}	G0 = {⟨?,?,?,?⟩}
1	⟨Sunny, Warm, Normal, Weak⟩ → Yes (+)	S1 = {⟨Sunny, Warm, Normal, Weak⟩}	G1 = {⟨?,?,?,?⟩}
2	⟨Sunny, Warm, High, Strong⟩ → Yes (+)	S2 = {⟨Sunny, Warm, ?, ?⟩}	G2 = {⟨?,?,?,?⟩}
3	⟨Rainy, Cold, High, Weak⟩ → No (−)	S3 = {⟨Sunny, Warm, ?, ?⟩} (unchanged; it does not cover the negative)	G3 = {⟨Sunny,?,?,?⟩, ⟨?,Warm,?,?⟩}
4	⟨Sunny, Cold, Normal, Weak⟩ → Yes (+)	S4 = {⟨Sunny, ?, ?, ?⟩}	G4 = {⟨Sunny,?,?,?⟩}
5	⟨Rainy, Warm, Normal, Strong⟩ → No (−)	S5 = {⟨Sunny, ?, ?, ?⟩}	G5 = {⟨Sunny,?,?,?⟩}

Detailed reasoning

- **Ex 1 (+):** S moves to the first positive example; G stays the most general.
- **Ex 2 (+):** Humidity (Normal vs High) and Wind (Weak vs Strong) differ, so both become "?". S2 = ⟨Sunny, Warm, ?, ?⟩. G covers the example, so no change.

- **Ex 3 (−):** G must exclude ⟨Rainy, Cold, High, Weak⟩. Minimal specialisations of ⟨?,?,?,?⟩ are ⟨Sunny,?,?,?⟩, ⟨?,Warm,?,?⟩, ⟨?,?,Normal,?⟩ and ⟨?,?,?,Strong⟩. The last two are discarded because they are not more general than S2 (S2 has "?" for Humidity and Wind). So G3 = {⟨Sunny,?,?,?⟩, ⟨?,Warm,?,?⟩}.

- **Ex 4 (+):** ⟨?,Warm,?,?⟩ fails to cover this positive (Cold) and is removed from G. S generalises Temperature (Warm vs Cold) to "?", giving S4 = ⟨Sunny,?,?,?⟩.
- **Ex 5 (−):** Both S4 and G4 classify Rainy as negative, so they are already consistent; no change.

Final Answer: S = {⟨Sunny, ?, ?, ?⟩} and G = {⟨Sunny, ?, ?, ?⟩}. Since S = G, the version space has converged to a single hypothesis: "A person enjoys the sport when the Weather is Sunny."

Question 5: Find-S Algorithm – Buys Product

Hypothesis form: ⟨Age, Income, Student, Credit Rating⟩. **Positive examples:** Ex 3 and Ex 4.

Negative (ignored): Ex 1 and Ex 2.

Initial hypothesis: h0 = ⟨∅, ∅ , ∅ , ∅ ⟩

Step	Example	Hypothesis after step	Reasoning
Ex 1, 2	Negative – ignored	h0 = ⟨∅,∅,∅,∅⟩	No change
Ex 3	⟨Middle, High, No, Fair⟩ (+)	h1 = ⟨Middle, High, No, Fair⟩	First positive: ∅ replaced by the example values
Ex 4	⟨Senior, Medium, No, Fair⟩ (+)	h2 = ⟨?, ?, No, Fair⟩	Middle≠Senior → ?; High≠Medium → ?; No and Fair match

Final Answer: Maximally specific hypothesis h = ⟨?, ?, No, Fair⟩, i.e. a customer buys the product if he/she is not a student and has a Fair credit rating (any Age and Income).

Check: Negative examples 1 and 2 are ⟨Young, High, No, Fair⟩ and ⟨Young, High, No, Excellent⟩. Ex 2 is correctly rejected (Excellent ≠ Fair), but Ex 1 is wrongly covered by h – a limitation of Find-S, which ignores negative examples.

Summary of Answers & Marking Guide

Q.No	Key result
1	8 blocks with sketch; supervised (spam filter) vs unsupervised (customer clustering)
2	Data, pre-processing, features, model; train/validate/evaluate; early warning & response
3	Final h = ⟨?, ?, ?, ?⟩
4	S = G = {⟨Sunny, ?, ?, ?⟩}
5	Final h = ⟨?, ?, No, Fair⟩

Rubric weightage applied: Conceptual Understanding 30%, Accuracy 20%, Application of Knowledge 20%, Completeness 15%, Clarity & Presentation 15%.